EVALUATION OF CONVENTIONAL, SERIAL, AND CHORD KEYBOARD OPTIONS FOR MAIL ENCODING

Rose Mae M. Richardson, Ronald U. Telson, Christopher G. Koch, and Susan T. Chrysler

Honeywell Inc.
Corporate Systems Development Division
Golden Valley, Minnesota

ABSTRACT

The motivation for the experiment was the integration of a work station display component and a suitable data entry device. Keying performance with three candi- date keyboards was evaluated: (1) a one-handed conventional calculator keyboard, (2) a two-handed 10-key serial keyboard, and (3) a two-handed 10-key chord keyboard. Subjects were trained to criteria on a numerical data entry task and subsequently tested for performance in encoding five-digit strings. Measures of encoding response time and encoding errors showed a substantial advantage in training time for the calculator keyboard and two-handed serial keyboard over the two-handed chord keyboard.

INTRODUCTION

The present study was carried out as part of a research program to study, design, and demonstrate an image processing system that will provide video images of mail pieces–including letters, flats, and irregular pieces and parcels–to operators for high-speed data entry. The experiment reported here involves the comparison of three candidate keyboard designs for ZIP Code encoding. These include a conventional one-handed calculator keyboard, a split, two-handed ergonomic keyboard used in sequential (serial) mode, and the split, two-handed ergonomic keyboard used in chord mode. The purpose of the study was to provide a basis for the evaluation of these keyboards, obtain performance data, and assess the generality of some earlier work carried out on keyboards of this type.

Although the conventional one-handed calculator configuration data entry keypad is an effective human-computer data entry device, a chord or multipress keyboard is often suggested as a potential alternative. The chord keyboard is generally configured with 10 keys, one for each finger of each hand. As the name implies, an input may involve any combination of the 10 keys held down simultaneously. There are 1023 possible combinations of from one to 10 keys. Some important advantages of the chord keyboard operating mode apply: there is no need for the hands or fingers to be moved from the home position, and a large number of codes can be entered with a single response.

Counterpoints to the effectiveness of the chord keyboard have been reported in a number of studies (Ratz and Ritchie, 1961; Siebel, 1962, 1963; Bartram and Lee, 1985), which evidence the complexity of different chord patterns and the relative shallowness of the learning curve. Average reaction times have ranged from 281 to 352 msec, with an error rate varying between 6 and 22 percent for the 31 chord patterns involving one hand (Siebel, 1962). Siebel (1963) reports continued improvement after 50,000 trials and a relative asymptote only after 75,000 responses. Furthermore, reaction times to the full 1023 set of alternatives averaged about 400 msec for two-handed chords, whereas those for one-handed chords were about 30 msec faster. In spite of differences among these pieces of research, close agreement can be found both for the relative difficulty of different chord patterns and for a stable pattern of difficulty within and between subjects.

Another keyboard configuration that has generated some interest (Bartram, Feggou and Lee, 1985) is the split-handed 10-key keyboard operated in serial mode. The key layouts on this keyboard can be physically identical to that of the chord keyboard, but the operation is sequential like conventional keyboards. In an evaluation of four different keyboard designs comprising both the chord and the serial layouts, Bartram *et al.* (1985) found the chord keyboard to be superior in terms of final speed, final accuracy, and rate of improvement. The study used a subset of 256 combinations for the chord keyboard and a total of 288 trials.

The pertinent questions we wanted to answer were:

(1) Can people perform a data input task for up to three numerals presented on a CRT at a rate of one numeral string per second?

(2) Do certain keyboard configurations and modes offer advantages in training time?

(3) Do certain keyboard configurations and modes evidence better relative performance after training to criteria?

METHOD

Our interest in the present study was the encoding of ZIP Code numerals, so the number of responses was limited to the digits 0 through 9. Numeral combinations for the chord keyboard were assigned according to the ten patterns with the smallest mean reaction times

from Siebel's (1962) work. The digit-response mapping for all keyboards is depicted in Figures 1, 2, and 3.

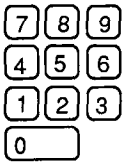

Figure 1. Key Layout for Calculator Keyboard

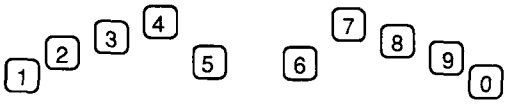

Figure 2. Key Layout for 10-key Serial Keyboard

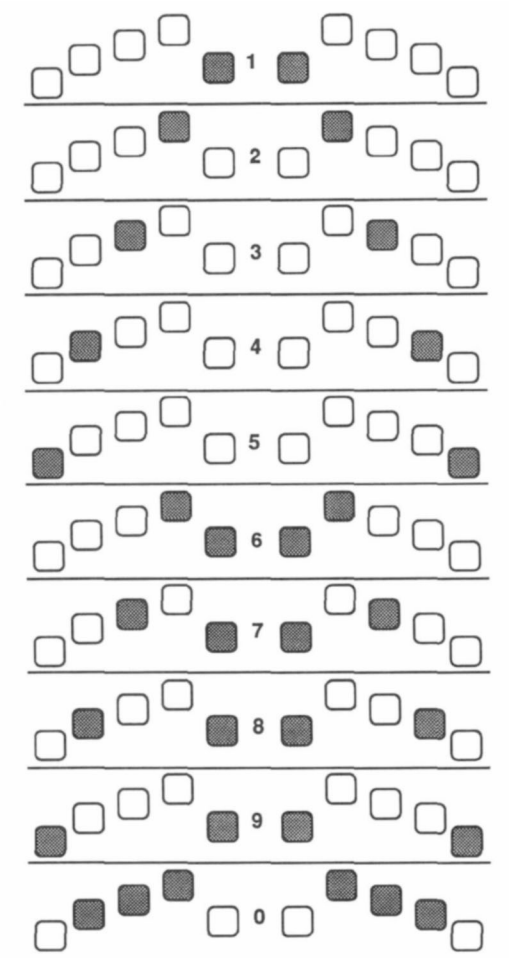

Figure 3. Key Layout for Chord Keyboard

Subjects

Twelve subjects, five females and seven males, obtained from an agency for office temporaries participated in this experiment. The mean age was 25 years ranging from 18 to 37. All subjects were tested for normal vision. They received weekly pay checks from the contracting agency.

Apparatus

Training and testing tasks were conducted on three dedicated ergonomic work stations equipped with one of the keyboards under study and a CRT for the display of stimuli. Stimulus generation and data recording were performed by an IBM PC/AT at each work station. The keyboard and CRT position and height were completely adjustable, and a fully adjustable ergonomic chair and footrest were provided.

The conventional one-handed keyboard used in this study was a "Touchstone 2" keypad equipped with a standard 10-key calculator keypad plus 25 other function and cursor-control keys. Subjects used only the calculator keypad portion of the keyboard. This keyboard is shown in Figure 4. The "5" key in the middle of the numeric keypad had a dimple to aid in finger positioning.

The two-handed keyboard shown in Figure 5 was provided by the British Post Office Research Centre. The split-handed 10-key ergonomic layout allowed subjects to place one finger over each key while retaining a comfortable hand position. This keyboard could be switched for use in either serial or chord data entry. In

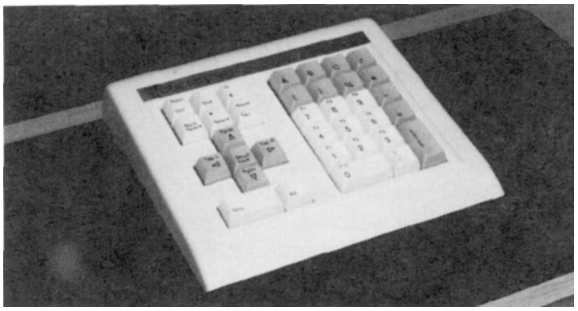

Figure 4. Conventional Calculator Keyboard

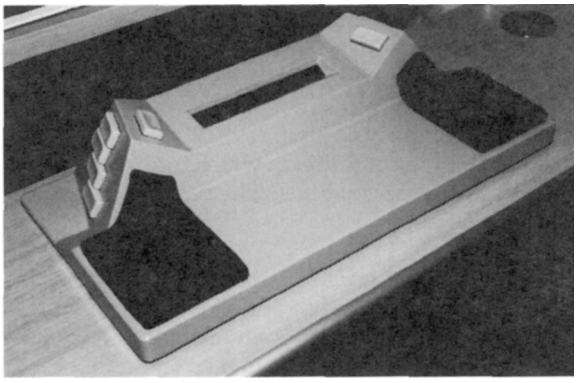

Figure 5. Keyboard for Serial or Chord Entry

chord mode, the keyboard acts as two separate mirror-imaged keyboards for left and right hands, allowing the simultaneous entry of two chords (that is, digits). For two such simultaneous chords, the left- hand keypad represents the first in the series.

Stimuli. The stimuli appeared in the center of a 13-inch (33 cm) color monitor equipped with an EGA board. The numerals were 18 by 18 pixels and presented in white on a black background. An entire numeral of the stimulus string could be enclosed within a square measuring 6 mm per side; the distance between numerals was about 1 mm. The numerals were in standard ASCII format. Stimuli were arranged in pseudorandom sequences, and the sequences differed daily.

During the training period, strings of one, two, or three numerals were used. In the testing phase, subjects were presented with five-digit strings. For the chord keyboard, the one-digit training stimuli were actually two identical digits (e.g., 22, 55) which were to be keyed simultaneously with both hands.

In the training phase, the subjects received auditory feedback (a 750 msec tone) signifying an erroneous response. No accuracy feedback was provided during testing.

Procedure. The subjects' task was to type a numeral string presented on the CRT as quickly and accurately as possible. In response to a stimulus string, the subject depressed the chord or sequence of keys corresponding to the digits displayed. Reaction time was measured from the onset of the stimulus pattern to the completion of the response. (For the calculator and serial keyboards, the response was considered to be complete when the key corresponding to the last digit of the stimulus was depressed. For the chord keyboard, the response was considered to be complete when the last key of the chord was released.) Each stimulus remained on the screen until the subject had typed the response. No "ENTER" or "RETURN" keypress was required. The next stimulus followed without an interstimulus delay.

Each subject was first exposed to the one-digit training condition, which terminated when for three consecutive blocks (300 trials per block), the subjects encoded at an average rate of 100 strings per minute (600 msec per string) with an accuracy of at least 95 percent in each block. The two-digit condition ensued with a criterion of 70 strings per minute (857 msec per string) with identical accuracy requirements. Training ended with the three-digit condition in which the criteria were a rate of 65 strings per minute (923 msec per string) and 95 percent accuracy.

Training sessions lasted 3 hours and 10 minutes, based on a work schedule of approximately 30 minutes of encoding, followed by a 10-minute break, followed by another 30 minutes of encoding, and so on, until the expiration of the session time.

After completing training, subjects returned for one day of testing on five-digit strings. These strings were presented in the same manner as described above but with no auditory feedback for incorrect responses. The testing lasted for 150 minutes, which was divided into five 30-minute blocks with a short rest break between blocks.

RESULTS

At the time of this writing, data collection has not been completed. Five subjects have completed training and testing (two on the calculator keyboard and three on the 10-key serial keyboard). An additional seven subjects have completed training through two digits.

Training Data

The training data for all twelve subjects on both one and two digits is presented here. Since all subjects were required to train until a set criterion was reached, the dependent measure of interest is number of blocks to criterion. The number of one- and two-digit blocks required to achieve performance criterion for each subject is shown in Figures 6 and 7, respectively. Significant differences were found between number of blocks to criterion on the 10-key chord compared to both the 10-key serial and the calculator keyboards. For one-digit training, a Kruskal-Wallis test resulted in a chi-square of 6.23, $p = 0.044$. For two-digit training, the chi-square was 6.54, $p = 0.038$. A Mann-Whitney test was then performed to detect any differences between the 10-key serial and the calculator keyboards. No differences were detected. For one-digit training, the test yielded a U of 10.0, $p = 1.0$; and for two-digit training, a U of 7.0, $p = 0.556$ was found. A Mann-Whitney test was also performed on the three-digit training completed by five of the subjects (two on the calculator keyboard and three on the 10-key serial keyboard). No three-digit training data was available for the chord keyboard. The number of three-digit blocks to criterion for each subject is shown in Figure 8. The test for three-digit training revealed no significant difference in training between the 10-key serial and the calculator keyboards, $U = 3.0$, $p = 1.0$.

Testing Data

As mentioned above, test data have been collected for the calculator keypad and the serial keyboard only. Mean reaction time and mean percent error were recorded for each block of 300 trials. The results are shown in Figures 9 and 10.

The mean reaction time and the mean error rate for each subject are shown in Figure 11. A Mann-Whitney test was performed and no significant differences were found between keyboard types. For the reaction time data $U = 1.0$, $p = 0.40$, and for the error rate data $U = 3.0$, $p = 1.0$. No difference was found for number of blocks completed in the testing period, $U = 1.0$, $p = 0.40$.

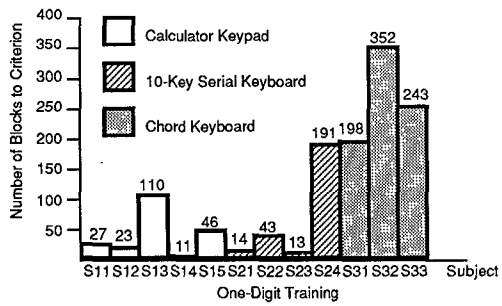

Figure 6. Blocks to Criteria for One-Digit Training

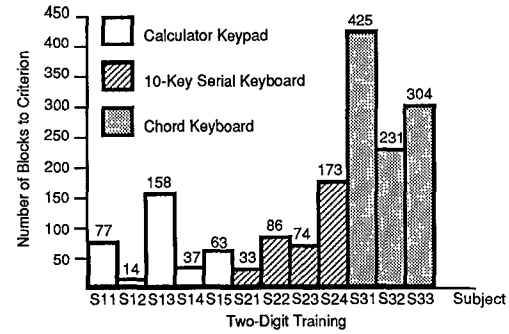

Figure 7. Blocks to Criteria for Two-Digit Training

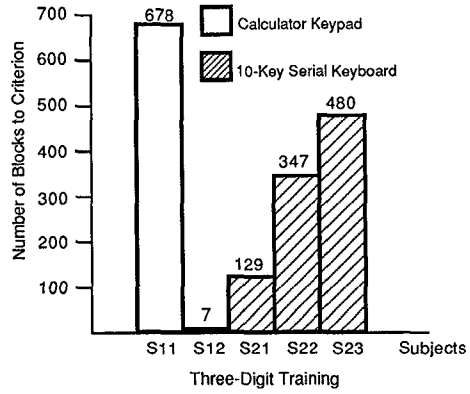

Figure 8. Blocks to Criteria for Three-Digit Training

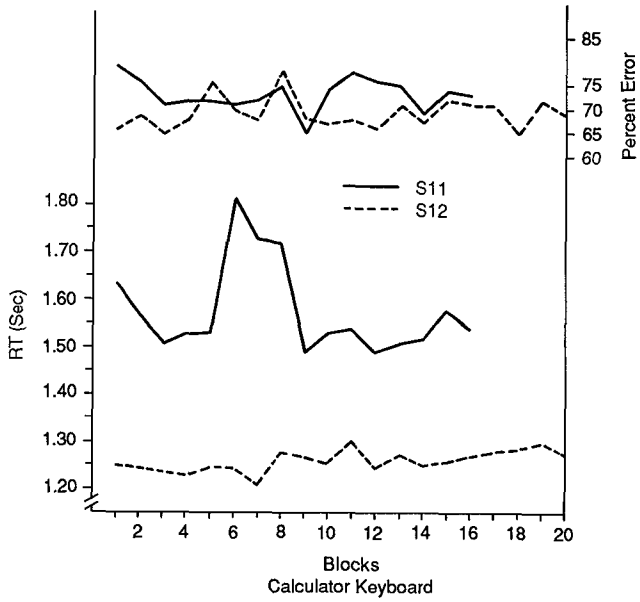

Figure 9. Mean Reaction Time and Mean Percent Error by Block by Subject for Calculator Keyboard Testing

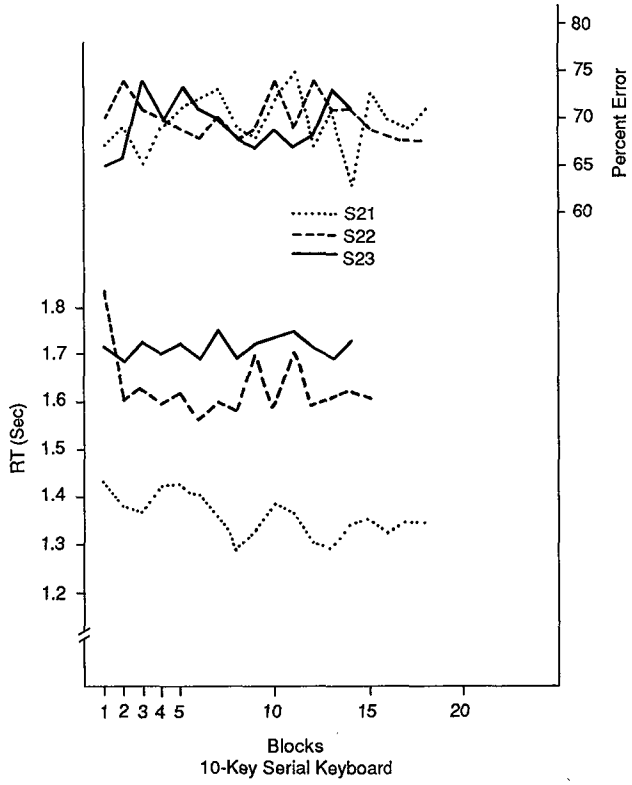

Figure 10. Mean Reaction Time and Mean Percent Error by Block by Subject for 10-Key Serial Keyboard Testing

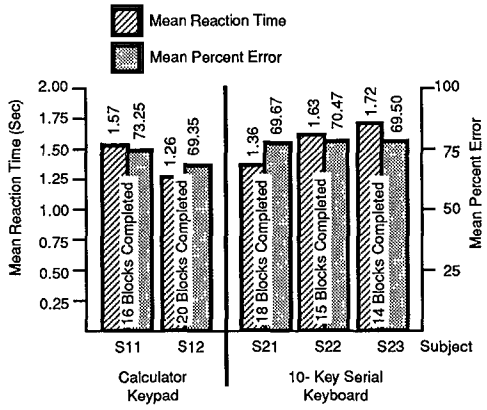

Figure 11. Mean Reaction Time and Mean Percent Error by Subject for Testing

DISCUSSION

The sorting of nonmachinable letters, flats, and irregular pieces and parcels is a labor-intensive operation at the U.S. Postal Service. The prospects for automation of materials handling of mail pieces, image acquisition, image processing and manipulation, bar code sorting, and optical character recognition of typed addresses is good. One of the hardest problems that remains is sorting mail pieces with addresses that cannot be read by machine. Human operators excel at this task, and this study of manual ZIP Code data entry is one critical element of the prototype semiautomated video encoding system. The goal for operator performance on the prototype system is machine-paced throughput capability of one mail piece per second per operator. Such a task involves maximum speed, accuracy, and versatility of encoding destination addresses with minimum stress, fatigue, and training for the operators.

One objective of the study is to ascertain that operators can perform the encoding task using video imagery presented at the prescribed rate. Results of the training for three keyboard configurations–one-handed calculator, two-handed 10-key serial, and two-handed 10-key chord–show that rate and accuracy criteria can be achieved. Operators reached target speeds of 600 msec, 857 msec, and 923 msec, each with 95 percent accuracy, for the one-digit encoding, two-digit encoding, and three-digit encoding tasks, respectively. Note that the stimulus materials designed to resemble ZIP Codes were standardized in numeral size, string orientation, string position, and numeral/background contrast. They were presented in a controlled environment at comfortable, properly adjusted ergonomic work stations. Consequently, we conclude that the legibility and readability of the numeral strings were nearly optimal.

Given that the desired throughput rate can be achieved, a second objective of the study is to determine whether one type of keyboard configuration offers advantages in training time. There is evidence for differences in training time among the keyboards. Fewer trials to criterion performance are associated with the calculator and serial keyboard configurations than the chord keyboard. In terms of training time for two-digit encoding, the mean time required for the calculator and serial keyboards is about 132 blocks of 300 trials (about 39,600 trials total). Assuming training averages 2 seconds per trial, required training time exclusive of rest breaks would be about 22 hours. The mean time required for criterion two-digit encoding with the chord keyboard is about 584 blocks (about 175,200 trials), with a training time of about 97 hours.

It is impossible to advance any conclusion about absolute performance effectiveness of the keyboards from the study because the training phase was not designed to achieve asymptotic performance. However, a third objective of the study is to determine if relative performance advantages can be expected for the calculator, serial, or chord keyboards given the amount of training that was provided. The preliminary data for the transfer of training task using five-digit encoding shows no difference in performance between the calculator and serial keyboards. In fact, the subjects performed more poorly than expected in the testing phase–the combined mean response time was about 1.5 seconds and mean error rate was about 70 percent. The high rate of error can be partially attributed to withholding feedback on accuracy. Also, the subjects must have interpreted that a trade-off of accuracy for speed in the testing phase was appropriate.

The additional work to be accomplished, once this experiment is completed, will address some other critical issues of the operator entry of mail sortation data:

- Requirement for command input by function key, such as "repeat," "clear," "reject," and "select," and how the keyboards can accommodate this.
- Incidence and effects of stress and fatigue on long-term encoding performance.
- Means for selecting a region of the CRT image (such as the destination address block) for subsequent enhancement and re-presentation.

REFERENCES

Bartram, D., Feggou, L. and Lee, M. (1985). An Evaluation of Four Different Keyboard Designs for Foreign Destination Coding Desks. Ergonomics Research Group, London.

Bartram, D. and Lee, M. (1985). A Pilot Investigation of the Ten-Key Chord Keyboard. Ergonomic Research Group, London.

Ratz, H.C. and Ritchie, D.K. (1961). Operator performance on a chord keyboard. *Journal of Applied Psychology*, 1961, *45*, 303-308.

Seibel, R. (1962). Performance on a five-finger chord keyboard. *Journal of Applied Psychology*, *46*, 165-169.

Seibel, R (1963). Discrimination reaction time for a 1,023-alternative task. *Journal of Experimental Psychology*, *66*, 215-226.

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